

CHIP DESIGN COMPETITION TECHNICAL SPECIFICATIONS

2026



VERSIONS

Version	Date	Description	Changes
1.0	02.01.2026	2026 Initial Version	
1.1	20.02.2026	Competition Calendar	Application Deadline
1.2	06.04.2026	Competition Calendar	PDR Results
1.3	27.04.2026	Competition Calendar and Microcontroller Category Changes	DDR Submission and Result Deadlines, [MICRO] TMR TIM_PRE register example is fixed, [MICRO] grammar fixes, [MICRO] UART_CFG[0] ambiguity resolved, [MICRO] A-3 notes are fixed.

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ABBREVIATIONS

AI:	Artificial Intelligence
AXI:	AMBA High-Speed Bus
AMBA:	Advanced Microcontroller Bus Architecture
BİLGEM:	Bilişim ve Bilgi Güvenliği İleri Teknolojiler Araştırma Merkezi
CMS:	Corporate Management System
CSR:	Control/Status Register
DC:	Direct Current
DDK:	Advisory and Evaluation Committee (Danışma ve Değerlendirme Kurulu)
DEF:	Design Exchange Format
DRC:	Design Rule Check
EDA:	Electronic Design Automation
FPGA:	Field Programmable Gate Array
GDS:	Graphic Data System
GPIO:	General-Purpose Input/Output
HDL:	Hardware Description Language
I2C:	Inter-Integrated Circuit
JTAG:	Joint Test Action Group
LVS:	Layout versus Schematic
MCU:	Microcontroller Unit
PDK:	Process Design Kit
PSRR:	Power Supply Rejection Ratio
QSPI:	Quad SPI
RISC:	Reduced Instruction Set Architecture
RO:	Read-Only
RTL:	Register Transfer Level
RW:	Read-Write
SPI:	Serial Peripheral Interface
SoC:	System-on-Chip
SV:	SystemVerilog
T3:	Turkish Technology Team
TT:	Typical-Typical
TÜBİTAK:	Türkiye Bilimsel ve Teknolojik Araştırma Kurumu
UART:	Universal Asynchronous Receiver Transmitter
UVM:	Universal Verification Methodology

1. PURPOSE

The Chip (Integrated Circuit) Design Competition, which was opened to university students for the first time in 2022 within the scope of TEKNOFEST, aims to raise awareness in the field of microelectronic technologies in associate, undergraduate and graduate (master) students and to provide students with knowledge on IC design. The competition is also organized to encourage teamwork and to lead the formation of competent human resources in this field by developing micro-electronic technologies with practical applications.

This document has been created to define all the rules and requirements of the Chip Design Competition, which is organized by the TÜBİTAK and Yongatek Microelectronics within the scope of TEKNOFEST Aviation, Space and Technology Festival (TEKNOFEST) Technology Competitions. The Analog Design category is organized by **TÜBİTAK**, and the Microcontroller Design category is organized by **Yongatek Microelectronics**.

2. COMPETITION SCHEDULE

Table 2-1. Competition Schedule

DESCRIPTION	DATE
Application Deadline for the Competition	15.03.2026*
Submission Deadline for Preliminary Design Report (PDR)	16.03.2026 - 17.00
Announcement of the Teams Passing the Preselection based on PDR Results	12.04.2026
QA Session	13-17 April 2026
Submission Deadline for Detail Design Report (DDR)	07.06.2026 - 23.59
Announcement of Finalist Teams	26.06.2026
QA Session	6-10 July 2026
Deadline for Finalizing the Design	31.07.2026 - 17.00
Final Presentations and Demos	August – September 2026

* This date is only valid for teams applying from abroad.

3. GENERAL INFORMATION ABOUT THE COMPETITION

The competition consists of two categories: Analog and Microcontroller Design. For the competition organized by TÜBİTAK, designing of SerDes analog front-end and bandgap voltage reference circuit are expected in the Analog Design category. In the Microcontroller Design category organized by Yongatek Microelectronics, it is expected to design and verify a microcontroller with various interface elements and memory units using a ready-made processor core.

What is expected from the teams that will participate in the competition is to prepare the preliminary design report, detailed design report, complete the design, prepare the design outputs and make a presentation/demo in the final for the hardware that meets the design criteria specified in this document. (A demo is expected for MCU)

Applications will be made through the official website of TEKNOFEST Technology Competitions (<http://www.teknofest.org>) until March 15, 2026.

3.1. Participation Rules

- Higher education (associate, undergraduate and master) students studying in Türkiye and abroad can participate in the competition.
- Participation in the competition can be done individually or as a team. Teams must consist of a maximum of 5 people (excluding the advisor).
- The number of graduate students (master) in the teams cannot exceed 2.
- A competitor can take part in teams in different categories, but not in different teams in the same category.
- Teams can only have 1 person as an advisor. It is not required to have an advisor. An advisor can advise only one team per category.
- For the competitors, approved student documents and for the advisors, approved documents showing that they are faculty members/assistants, research assistants, PhD students or PhD graduates must be submitted with the Detail Design Report.
- Teams can be formed from a single school or as a mixed team with one or more higher education students coming together.
- Teams will be allowed to participate in each category with only one design. Multiple versions cannot be submitted for the same category.
- Competitors who have completed the team formation process must apply to the competition in accordance with their project.
- Between the application dates, the team captain/advisor registers through the system, makes the correct and complete registration of the advisor and/or team captain/team members, if any, and sends an invitation to the advisor and members' e-mails, if any. The member to whom the invitation is sent, logs in to the Application system, accepts the invitation from the "My Team Information" section and the registration is completed. Otherwise, the registration will not be completed.
- All necessary processes within the scope of the competition (Application, Report Submission, Announcement of the Report Results, Financial Support Application, Objection, Member Addition/Removal, etc.) are done through the CMS system. Teams are required to follow their processes through the CMS system.
- Adding/removing members can be done until the Detail Design Report Submission date.
- During the competition process, the processes of applying through the CMS, uploading reports and filling out forms are under the authority of the team captain and/or the advisor, and the competition processes are managed through these people.
- Transportation and accommodation support to be provided to the finalist teams is limited. The number of people to be supported is 3 people (including the advisor) per team and the TEKNOFEST Competitions Committee has the right to make changes.
- TEKNOFEST Competitions Committee has the authority to limit the number of members in the festival area. In case of restrictions, the committee will inform.

- Throughout the competition process, education level at the time of application will be considered.
- Applications of TÜBİTAK and Yongatek Microelectronics employees will not be evaluated.

3.2. Reports and Presentation

- The format in the **GENERAL RULES** section should be followed for references from previous year's reports. Other than that, IEEE format should be followed for references from other sources.
- Two reports, a preliminary design report and a detailed design report, will be submitted. The technical information required in the relevant headings of the reports should be conveyed in detail.
- Report templates are published on the [competition website](#).
- Reports can be prepared in Turkish or English. It is mandatory to use a single language in a report.
- The reports are uploaded to the application system in PDF format with a maximum size of 60 MB until 05.00 pm in the specified deadlines. Otherwise, the team will be eliminated from the competition.
- The reports should be prepared in A4 format, 11-point, Calibri font, with a line spacing of 1.15 and 2.5 cm at the bottom, top and sides. The preliminary design report should have a maximum of 9 pages, and the detailed design report should have a maximum of 30 pages (including the cover page, pictures, tables, references).
- The report is evaluated within the scope of scientific and technical criteria specified by DDK members who are experts in their fields. Teams with a score below what is determined by DDK are eliminated from the race.
- Teams with successful reports are announced on the <http://www.teknofest.org> page.
- In case of an ethical violation such as plagiarism, copying etc., the report will not be evaluated, and the corresponding team will be eliminated from the competition. This rule is applied for the application of the same university/club/society etc. regardless of category/race.

3.2.1. Preliminary Design Report (PDR)

Preliminary design report is expected to include the following sections:

Block diagram of the overall design, target performance summary table, system level importance of targeted performance measures, schematics of the possible critical circuit core, verification plan and methods, brief description of the circuit and design techniques to be used, applicable references (e.g., an article that inspires your design, all the open-source blocks you'll reuse, etc.). The design structure given in the PDR can be changed later, provided that change reasons are explained.

The up-to-date report template will be shared on the [competition website](#).

In order to proceed to the next stage within the scope of this competition, the preliminary design report must be submitted and deemed successful.

3.2.2. Detailed Design Report (DDR)

In the Analog Design category detailed design report, it is expected that the schematic level design will be completed and reported with the results.

The Microcontroller Unit (MCU) category detailed design report should include topics such as how the requirements presented in the specification are realized, block diagrams, simulation and synthesis results. The report shall also include the current verification status of the project; including the verification/test plans, test status, coverage reports and screenshots meanwhile addressing the artefacts in the Git repository.

The up-to-date report template will be shared on the [competition website](#).

Teams that have passed to the DDR stage are obliged to submit their reports on the date specified in **Table 2-1**.

According to the DDR results, the teams that will participate in the final evaluation will be announced on the date specified in **Table 2-1**.

3.2.3. Final Evaluation Presentation and Design Outputs

Final evaluation presentation and design outputs (schematic, RTL, GDSII etc.) must be submitted by the date specified as “Deadline for Finalizing the Design” in **Table 2-1**. Changes made to GitHub repositories after 11.59 pm on the relevant date will not be taken into consideration. The presentations should be uploaded on the same date until 05.00 pm. Presentations in the competition area will be made through presentation files uploaded to the system. Using the design outputs, the results in the presentation will be verified and the non-conforming groups will be eliminated. For this reason, the design outputs must be up-to-date and compatible with the presentation. Details of the design outputs are given in **Design Outputs** section.

3.3. Evaluation

Teams that pass the PDR stage will be entitled to participate in the DDR stage. PDR and DDR reports will be evaluated in accordance with the template to be announced on the [competition website](#). Teams that pass the DDR stage will be eligible to participate in the presentation and demo evaluation. Presentation evaluation will be made by the competition jury according to the oral presentation and the answers to the questions of the evaluators. Teams that are not in the competition area, although they are entitled to participate in the presentation evaluation, are considered withdrawn from the competition. The first, second, and third place teams will be determined by the weighted points from all stages.

3.3.1. Analog Design Category

Scoring will be made out of 100 points and reports, presentations, and demos will constitute the entire score. Reports are 40% of the total score and the remaining 60% will be presentation and performance criteria. Evaluation will be made through post- layout simulations in the Analog Design category.

The total score that can be obtained at the end of the competition will be a maximum of 100 points and the calculation will be made as follows.

Total Score = (0.15 * PDR Points) + (0.25 * DDR Points) + (0.35 * Presentation Points) + (0.25 * Performance Criteria Points)

The Performance Criteria Scores will be determined considering the eye opening after the receiver front-end circuit, the total area, and the power consumption.

3.3.2. Microcontroller Design Category

The system, which will be designed within the scope of the Microcontroller Design category, will consist of two separate parts: FPGA and physical design realization. The competitors will first create a design on an FPGA development board that will fulfill the tasks given in the Microcontroller section of the Microcontroller Design Category under the title of Design Requirements and run this design on the FPGA card. In the next stage, they will create GDSII outputs in the form to be sent to production with the IC design software that will be provided for physical design. Detailed information about the IC physical design will be clarified in the future.

Technical information about what needs to be done for the FPGA and Physical Design stages is defined in the Microcontroller Design Category heading under the section **TECHNICAL RULES**.

The total score that can be obtained at the end of the competition will be a maximum of 100 points and the calculation will be made as follows.

Total Points = (0.1 * PDR Points) + (0.15 * DDR Points) + (0.05 * Presentation Points) + (0.1 * Demo Points) + (0.4 * Design and Verification Points) + (0.2 * IC Flow Points)

Table 3-1 contains the outline of the scoring breakdown.

Table 3-1. Microcontroller category scoring breakdown

Points Category	Points Subcategory
PDR Points	Formatting
PDR Points	Flow of information
PDR Points	Rationale behind the decisions
DDR Points	Formatting
DDR Points	Flow of information
DDR Points	Rationale behind the decisions
DDR Points	Analyses of the results
Presentation Points	Timeliness regarding the presentation clock
Presentation Points	Presentation technique
Presentation Points	Smooth and adequate presentation of the outputs
Presentation Points	Clear explanation and analyses of the missing items
Presentation Points	Adequacy of the answers provided
Demo Points	FPGA demo
Demo Points	Obtaining the server-side outputs at-will
Demo Points	Adequacy of the answers provided
Design & Verification Points	Peripherals
Design & Verification Points	AI accelerator implementation
Design & Verification Points	System integration
Design & Verification Points	Compliance with the requirements
Design & Verification Points	Successful boot
Design & Verification Points	Successful program run
Design & Verification Points	System performance
Design & Verification Points	Verification and test plan
Design & Verification Points	Core verification
Design & Verification Points	Accelerator verification
Design & Verification Points	System-level verification

Design & Verification Points	Scope of the verification plan, and the cover rate of the defined scope
IC Flow Points	Static design checks (CDC, RDC, STA, vb.)
IC Flow Points	Synthesis
IC Flow Points	P&R
IC Flow Points	LVS, DRC, etc.
IC Flow Points	GDS outputs

3.4. Contact

For questions about the competition, you can join the [competition group](#) from the Chip Design Competition [page](#) on the TEKNOFEST [website](#). It is the responsibility of the competitors to actively follow this group and at least 1 person from each team to follow the announcements, questions and answers in this group as a member. TEKNOFEST Competitions Committee is not responsible for the inability of the teams to access up-to-date information that may arise as a result of not following the specified e-mail group.

Questions about the organizational parts of the competition should be sent via iletisim@teknofest.org e-mail address. It is important that your questions are sent through the correct channels above in order to respond quickly to the questions asked.

4. TECHNICAL RULES

4.1. Design Environment and Technology

In the Analog Design category, commercial-grade EDA tools (vendors like Synopsys, Cadence, Siemens) or an open-source EDA tools and it's appropriate PDK will be used. Version information for the PDK and the tools will be shared in a later announcement. Teams are required to design using the specified versions. If an open access EDA tool is used, the designs should be updated in a special repository opened on GitHub and shared with DDK before the delivery of the detailed design report and final presentation. If a commercial EDA tool is used, there will be no need for a repository opened on GitHub.

In the Microcontroller Design category, Verilog, SystemVerilog or VHDL hardware design language (HDL) must be used. The design that is shared with the competition evaluation board must be updated at least every 20 days in a special repository created on GitHub. Design contents that shall be uploaded to GitHub will be determined by DDK. During the chip design phase, commercial EDA tools (such as Synopsys, Cadence, Siemens) and PDK suitable for these tools, or open-source EDA tools and an appropriate PDK will be used will be used. Details will be shared by DDK at a later date.

The Competitors must provide documented and well-defined automation scripts (Python, Shell, Makefile, etc.) alongside their final design outputs in their GitHub repositories. What to use these scripts for and how to use them shall be commented within the scripts and documented within READMEs (txt, md, rst, pdf, etc.). The juries shall be able to perform all the flow stages such as simulation, synthesis, and implementation. The final results reported in DDR and presentations shall be the same as the ones being obtained through these scripts since the jury shall evaluate the final designs using these scripts. Therefore, all the inputs, outputs, explanations, script usages for multiple-script flows shall be explained clearly and

concisely. All the Competitors' design flows and jury evaluations will be executed on the same server environment with the same program versions.

The UVM library and SystemVerilog HDL should be used for {AXI or AXI-Lite} interface verification of the peripherals and the accelerator. When verifying the processor core defined in the specification that will be used in the SoC, the test programs should be shown to be working correctly by comparing them with the Spike ISS results.

GitHub repositories of the participating teams shall be shared with "yongatek-teknofest" account for the Microcontroller Design category. Competitors are required to share their GitHub repositories with the relevant accounts from the date to be determined later. Otherwise, they will lose points as stated in the scoring section. The start date of the regular repository update obligation will be announced later.

4.2. Design Requirements

4.2.1. Analog Design Category

The increasing data communication demands of our era are driving communication systems toward higher speeds and low-power, compact, and reliable solutions. Especially in applications such as **data centers, artificial intelligence accelerators, high-performance processors, 5G infrastructure, and high-resolution image processing systems**, the need for data transfer rates of tens of gigabits per second has emerged. These requirements make it necessary to develop highly efficient **SerDes (Serializer/Deserializer)** architectures capable of meeting the increasing speed and performance demands.

Figure 4-1 shows a typical **wireline transceiver** architecture. Signals arriving from high-speed differential lines (such as PCI Express) are terminated with the appropriate impedance in the receiver block and captured by the analog front-end. In this stage, equalization and amplification are applied, and the signal is forwarded to the decision circuit (slicer). The clock signal for the decision circuit is generated with the help of a CDR. On the transmitter side, digital data is delivered to the differential lines at the appropriate output amplitude through the driver stage.

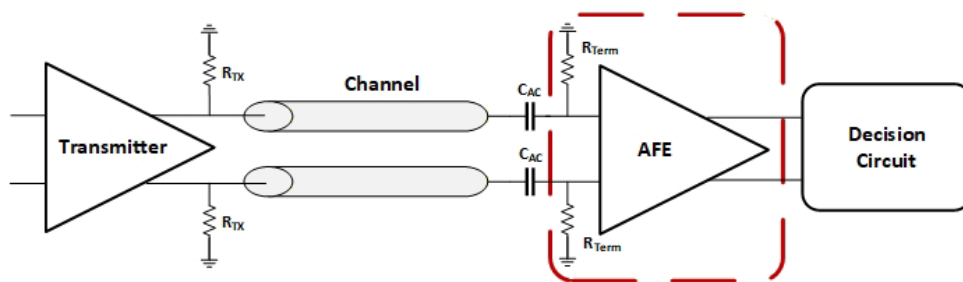


Figure 4-1. Wireline Transceiver Diagram

PCI Express (PCIe) is a serial communication-based interface standard used for **high-speed data transfer** in modern computers. Replacing traditional parallel PCI buses, PCIe transmits data serially over differential pairs called **lanes**, enabling higher data rates with fewer pins. The PCIe standard provides reliable and low-latency communication between hardware components such as **processors, graphics cards, NVMe-based storage devices, FPGAs, and high-speed network interfaces**.

The **PCIe 3.0 standard** is the third generation of this architecture and offers a raw data rate of **8 gigabits per second (Gbps)** per lane. In this competition, participants are expected to design an **analog front-end (AFE) supporting the PCIe 3.0 standard**. The design should include the essential analog blocks on both the transmitter and receiver sides and must meet key criteria such as **input/output impedance matching, frequency response, and output eye opening**. In addition, a **bandgap reference circuit**, which is required for biasing the system, must be designed.

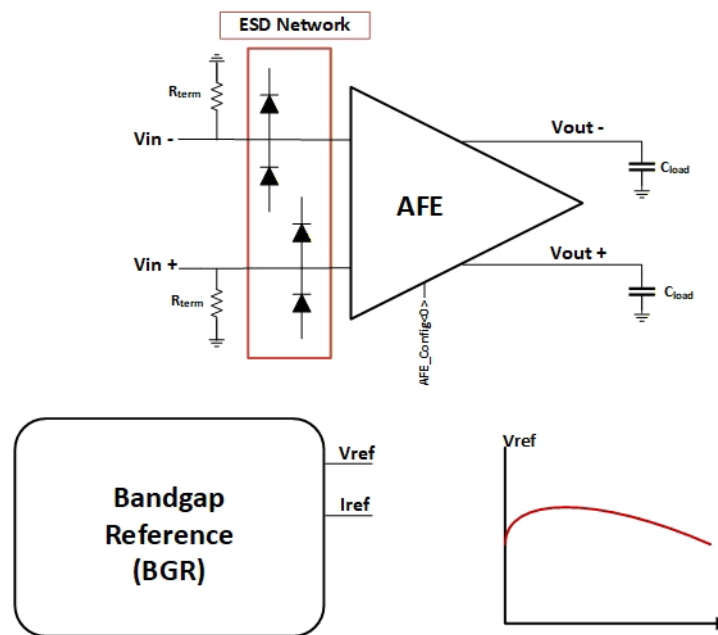


Figure 4-2. Block Diagram of Analog Front-end and Bandgap Reference Voltage Circuits

Block level and general design requirements are given below in detail.

4.2.1.1. General Requirements

- The design environment and technology specified in Section 4.1 must be complied with.
- All circuits should be powered by only one ideal voltage source of 1.8 V.
- All circuit elements should be used from the process library, ideal elements should not be used.
- Simulations should be made at the TT (Typical-Typical) process corner and at 27°C. The 27°C temperature is the nominal temperature at which the entire design will operate. Only simulations of the bandgap reference voltage circuit should cover the temperature range of -40°C to 125°C.
- Verify that the following requirements are met by post-layout simulations.
- Schematic and post-layout simulation results should be presented together.
- Except for start-up circuits, all MOS transistors must operate in the saturation region. (Applicable to band-gap reference circuit)
- Stability analysis should be performed when necessary.
- The signal flow is differential throughout the system.

- The channel should have a loss of 20 dB and 10 dB at 4 GHz, in two different conditions. It will be AC-coupled with a series 250 nF capacitance (C_{AC}). The channel models will be provided by the DDK during the DTR phase.
- The common-mode voltage of the transmitter must be half of the supply voltage.
- The transmitter swing should be modeled as 800 mV_{pp,diff}. The rise and fall time, between 10% and 90%, must be 20 ps.
- The transmitter output impedance will be modeled as single-ended 50 Ω or differential 100 Ω (R_{TX}).
- The transmitter data will be provided in PRBS form by the DDK at the DTR phase.

4.2.1.2. Analog Front-End Requirements

- The ESD protection circuit elements, which will be provided by the DDK during the DTR stage, must be included in the front-end circuitry.
- The circuit must have 50-ohm termination (R_{term}).
- A 20 fF load capacitance (C_{load}) must be connected at each differential node to represent the decision circuit.
- When viewed from the receiver input ports, the S_{11} value should be better than 10 dB over the range of 0 Hz to 4 GHz (100-MHz steps), with respect to either a single-ended 50-ohm reference or a differential 100-ohm reference (whichever is chosen).
- At the decision circuit input, the eye height must be 250 mV_{pp,diff} $\pm$ 10% for both channel loss conditions. Digital-controlled adjustment can be performed for the two different channel conditions (AFE_Config<0>).
- At the decision circuit input, the eye width must be at least 0.35 UI for both channel loss conditions. One UI is 125 ps for 8 Gbps.

4.2.1.3. Bandgap Reference Voltage Circuit Requirements

- A 1.25 V reference voltage should be generated as shown in Figure 2 for the purpose of validating the bandgap reference circuit.
- The reference voltage should vary by a maximum of 15 ppm/ $^{\circ}$ C, covering the temperature range of -40 $^{\circ}$ C to 125 $^{\circ}$ C.
- The circuit should generate a reference current of 16 μ A. These reference currents will be used as biasing currents in other circuits.
- For 1.8 $\pm$ 10% supply voltage, the 1.25 V reference voltage should not change and the transistors should protect their nominal operating regions.
- A start-up circuit should be utilized to avoid circuit startup problems.
- The PSRR value should be below 0 dB in the 1 Hz-10 GHz range.

4.2.2. Microcontroller Design Category

4.2.2.1. General Requirements

Two solutions will be realized under the microcontroller category: The first will work on FPGA (FPGA prototyping), the second will be to create production files with the physical design tool (IC Design).

The microcontroller to be designed by the competitors will use the CV32E40P RISC-V core IP with a single-core, 32-bit 4-stage pipeline, which is maintained by the "OpenHW Group" and available as open source in the GitHub environment. The GitHub repo link of the relevant processor core is given below:

<https://github.com/openhwgroup/cv32e40p>

The documentation link for the CV32E40P processor core is given below:

<https://docs.openhwgroup.org/projects/cv32e40p-user-manual/en/latest/index.html>

Which release and commit version, and which configuration of the IP will be used within the scope of the competition, that is, which commands it will support or not, will be clarified by DDK at a later date.

The microcontroller shall have an AI (artificial intelligence) hardware accelerator. In order to load AI accelerator input data, a dedicated UART interface shall be used to drive the data into the accelerator. Details regarding the data transfers and the AI accelerator can be examined from section **A-1 AI Accelerator Requirements**. The AI accelerator shall be designed by the competitors based on the requirements provided in the corresponding section. AI accelerators are expected to be evaluated and assessed in terms of performance, which shall be based on data/clock cycles and data/second based on the synthesis frequency, and mixed workload execution time compared with the standalone core.

The microcontroller shall have the following peripherals:

- 1x GPIO (32 pin I/O)
- 1x Timer
- 2x UART (one general-purpose and one for AI accelerator data stream purposes)
- 1x I2C Master
- 1x QSPI Master
- 1x JTAG (Optional)

Register definitions and other details regarding peripheral units are defined under the section **A-2 Peripheral Registers**. The competitors can design peripheral designs themselves, or they can use open-source projects by referencing them.

Validation of the CV32E40P RISC-V processor core is expected to be done with an instruction set emulator (ISP) (e.g. Spike ISS). Spike ISS GitHub repo link:

<https://github.com/riscv-software-src/riscv-isa-sim>

As part of microcontroller verification, the verification of {AXI or AXI-Lite} interfaces of the peripherals and the AI accelerator is expected to be done using SystemVerilog HDL and Universal Verification Methodology (UVM). In the verification work done, regression and coverage result reporting would increase the verification score. The environment in which UVM will be used (Synopsys, Cadence, Mentor) will be clarified by DDK at a later date.

It is not necessary to use UVM/SV for further purposes than protocol checking. However, teams are encouraged to use those agents for setting up better self-checking testbench environments as these agents would packet all the AXI transactions into meaningfully grouped data that could be utilized for any purpose. There may be a variable amount of {AXI or AXI-Lite} components on the designs.

Verification of the components can be performed with non-UVM methods and methodologies. However, all peripherals, AI accelerator, and the MCU top module must be verified with at least some directed testing on system-level. Section **A-3 Verification Methods** can be referenced for details regarding the verification.

Core will contain 8 kB instruction and 8 kB data memory, meanwhile the AI accelerator will contain 30 kB memory. (Accelerator memory may be a simulation model rather than an SRAM macro for initial development stages. Details regarding PDK and macros will be announced by DDK at a later date.)

Addressing of peripherals and memories will be defined and designed by the competitors.

When the system is powered on, it will boot from a non-volatile memory (QSPI Flash Memory) via the QSPI master interface. The bootloader code will be located in a small ROM (e.g. 512 Bytes or 1 kB) as a piece of code that cannot be changed.

When the system is powered on, the Bootloader code will be run first. This piece of code will read the program to be run from Flash memory via the QSPI Master interface and write it to the instruction memory. Afterwards, it will transfer the execution to the instruction memory.

The system shall be capable of handling switches between different peripherals or components without requiring any system resets. So, the system shall be capable of switching to a “ready state” after usage of each component by itself.

Optionally, a debug module connected to the JTAG interface will be designed. The debug module will be connected to the "debug" port of the CV32E40P processor core and the program running on the MCU can be debugged through this debug module.

4.2.2.2. Design Flow

This section provides an example design flow to answer some of the question marks regarding the specifications and provides an example roadmap to the Competitors.

- Since the AI accelerator shall use AXI-Lite for configuration the chip is expected to have an {AXI or AXI-Lite} interconnect. These interconnects shall be AXI4-compatible.
- Since the AI accelerator would have its own memory region, there is no need to have multiple masters for the same memory.
- AI accelerator shall raise a CPU interrupt when its execution is complete, and the classification result is determined. Once the CPU receives the interrupt, it shall execute the necessary steps, in other words, interrupt service routine (such as

reading specific registers and initiating a UART TX operation) to receive the classification result, and print out the results through UART interface before returning to normal operation.

- If a system-level testbench environment were to be set up in parallel to the design flow, each component can be verified as the integration activities go on. Besides, since only tests can actually verify the successful design and/or integration, each component shall be tested on the system to some extent.
- The provided model is a standalone software-based implementation of its own; thus, it can be used as a reference model for the accelerator IP block that would be integrated.
- Competitors are free with their ways of implementing the AI accelerator memory mapping.
- For the performance evaluation of the accelerators, the initial data read, and the final data write intervals within the simulations may be utilized.
- Competitors should prepare verification and test plans to define the scope of the verification, define the methods (test scenarios, coverage, assertion, etc.) to cover their verification requirements and report their verification completeness/progress, which would increase the verification scoring. The scope of the defined verification and test plan is another metric that would be evaluated besides the verification completeness rate.

Design Outputs

For the **Analog Design category**, the expected design outputs from the analog front-end and bandgap reference voltage circuits are as follows:

- GDSII (.gds) (GDS: Graphic Design System): The industry standard layout file format used for submitting the design to production. It will be used for layout evaluation and physical verifications (DRC/LVS).
- Design and test environment schematics: These are the files created by EDA tool Schematic Editor that contain the transistor-level designs, and the test environment related to these designs. All schematic files used in the project should be included in the printouts to validate my designs.
- Schematic and post-layout transistor level netlist (.spice/.scs/.l): The transistor level netlist is the file showing the PDK elements in the design and their connections with each other. In the post-deployment netlist, parasitic elements are also included in this file. This file will be used for pre- and post-layout performance verifications.
- Layout output files: EDA tool Layout Editor output file. It is requested for the control of the layout. The final design layout must be ready to be sent to MPW, including wire bond PADs and all chip-level signoff tests completed. (IP-level or block-level layouts will not be considered sufficient.)

- Physical Verification Reports: It consists of reports containing the results of physical verification tests (DRC/LVS).

Teams are required to upload their work to a remote repository or server location determined by DDK by a date to be specified later. An announcement about the upload format will be shared with the teams.

For the **Microcontroller Design category**, different outputs are expected for FPGA and physical design flows. The outputs are expected to be in the GitHub account, which the teams will share with the Microcontroller Design category regulatory company.

FPGA Flow

- All RTL Verilog/SV design codes
- All testbench Verilog/SV/VHDL codes,
- All testbench Verilog/SV/VHDL codes,
- Successful (error free) synthesis result report,
- Successful (error free) static timing analysis report,
- Successful (error free) implementation (Place & Route) result report,
- Bitstream that can be loaded into the FPGA.

IC Physical Design Flow

- All RTL Verilog/SV/VHDL design codes,
- All testbench Verilog/SV/VHDL codes,
- Successful completion of physical design flow,
- Production ready LEF, DEF, GDSII file outputs of the design,
- Successful completion of physical design signoff tests. (LVS, DRC, ERC, etc.),
- Automation scripts regarding the flow.

5. AWARD

As a result of the evaluation, the teams that pass the report stages and reach the finals in their own category, meet the award criteria and are ranked in the final evaluation, will be awarded a monetary award as indicated in **Table 5-1** and **Table 5-2. Categories will be evaluated and awarded separately.** First, second and third place prizes will be divided equally according to the total number of team members (all members registered in the system) and will be deposited into the bank account specified by each person. The relevant category organizer will determine whether the teams that cannot meet the minimum success criteria for the award ranking in the competition categories can receive an honorable mention award or how much they can receive.

The prizes in the table below show the total amount that will be awarded to the teams that are eligible to receive the prize, no individual prizes will be awarded. The first, second and third prizes will be divided equally according to the total number of team members (excluding the advisor) and will be deposited into the bank account specified by each person. Team advisors who are entitled to receive an award cannot benefit from the first, second and third prize amounts below, the awards to be given to the advisors are also stated in the table below.

Table 5-1. Analog Category Awards

RANK	AWARD AMOUNT	ADVISOR
First	200.000,00 ₺	20.000,00 ₺
Second	175.000,00 ₺	15.000,00 ₺
Third	150.000,00 ₺	12.000,00 ₺

Table 5-2. Microcontroller Category Awards

RANK	AWARD AMOUNT	ADVISOR
First	200.000,00 ₺	20.000,00 ₺
Second	175.000,00 ₺	15.000,00 ₺
Third	150.000,00 ₺	12.000,00 ₺

5.1. Minimum Success Criteria for Award Ranking in Analog Category

Competitors must meet the following specifications at post-layout simulations in order to enter the award ranking.

- The analog front-end and bandgap reference circuit must function properly.
- The output eye opening requirements of the analog front-end must be met for post-layout.
- The temperature coefficient and PSRR requirements of the bandgap reference circuit must be met for post-layout.

Also, the competitors must have passed the presentation stage and be present in the festival area on the day of the event.

5.2. Minimum Success Criteria for Award Ranking in Microcontroller Category

The minimum success criteria required for the Competitors to be eligible for an award are given below.

- Demonstration that the microcontroller successfully carries out the test scenarios given by the DDK on the FPGA development board.
- Successful execution of at least one self-checking system-level test that executes the core boot flow, and successful programming and usage of an IP block.
- Verifying the {AXI or AXI-Lite} interfaces of the peripherals and the AI accelerator through AXI agents that do protocol checks on the corresponding buses. The minimum criteria here is to have the protocol checkers integrated.
- Verifying the AI accelerator through at least one test scenario.
- Preparing GDSII outputs ready for production by successfully completing the physical design flow.

6. GENERAL RULES

[Click here](#) to access the General Rules booklet which is valid for the competition.

7. CODES OF CONDUCT

[Click here](#) to access the Code of Ethics booklet that is valid for the competition.

8. STATEMENT OF RESPONSIBILITY

T3 Foundation and TEKNOFEST are not responsible for any product delivered by the competitors or any injury or damage caused by the competitor. T3 Foundation and organization officials are not responsible for the damages caused by the competitors to third parties. T3 Foundation and TEKNOFEST are not responsible for ensuring that the teams prepare and implement their systems within the framework of the laws of the Republic of Türkiye.

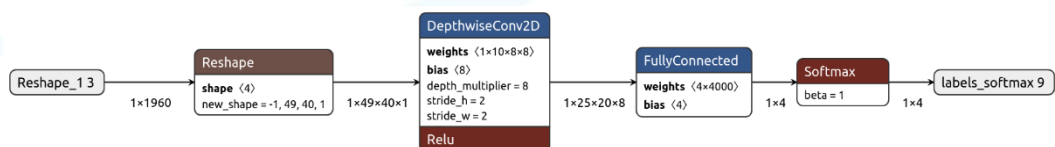
Türkiye Technology Team Foundation reserves the right to make any changes in this specification.

9. APPENDIX

A-1 AI Accelerator Requirements

The AI accelerator shall have an accelerator architecture with convolution model implementation inside. The AI accelerator requirements are defined below:

- The AI accelerator shall implement the Micro Speech model from TFLite on RTL level.
 - https://github.com/tensorflow/tflite-micro/tree/main/tensorflow/lite/micro/examples/micro_speech<https://github.com/tensorflow/tflite-micro/tree/main/tensorflow/lite/micro/kernels> Micro Speech model uses the Tiny Conv architecture that consists of one 2-D convolution layer, one Fully Connected layer, and input and output layers. Model takes the preprocessed audio inputs and classifies the inputs as the word “yes”, the word “no”, unknown word, or silence. Model will be explained in this section, but official GitHub project can be referred for more information and hands-on trials. https://github.com/tensorflow/tflitemicro/tree/main/tensorflow/lite/micro/examples/micro_speech
 - Model visualization and brief explanation of the architecture layers are given below.



- The model begins by accepting a flat input vector of 1960 values, which represents a one-second audio clip pre-processed into a spectrogram. The first operation is a Reshape layer that reorganizes this flat vector into a 3D tensor of 49 time steps, 40 frequency bins, and 1 channel (49x40x1). This reconstruction is essential because it restores the spatial structure of

the audio data, treating it like a single-channel grayscale image, which allows the subsequent convolutional layer to detect two-dimensional patterns within the sound frequencies over time.

- The core feature extraction is performed by a DepthwiseConv2D layer. This layer applies 8 distinct filters (kernels), each with a size of 10x8, across the input array. These kernels are not constant; weight values are also updated in the training process. By using a depth multiplier of 8, the layer expands the feature space, learning to recognize eight specific audio characteristics. The layer also utilizes strides of 2 in both height and width, which effectively down-samples the data, reducing the spatial dimensions from 49x40 to 25x20. A fused Rectified Linear Unit (Relu) activation function is applied immediately after the convolution to introduce non-linearity, ensuring the model can learn complex boundaries by zeroing out negative values.
- Following the convolution, the resulting 25x20x8 feature map is flattened into a vector of 4000 elements and passed to a FullyConnected (dense) layer. This layer performs matrix multiplication using trained weights and biases to compress the extracted features into 4 raw scores, representing the model's confidence for each class.
- Finally, a Softmax layer normalizes these raw scores into probabilities that sum to 1. The result is a final output vector indicating the likelihood that the audio clip belongs to one of the four target categories: Silence, Unknown, Yes, or No.
- Model is originally software implemented as an example TFLite project, using base functions in the <https://github.com/tensorflow/tflite-micro/tree/main/tensorflow/lite/micro/kernels>. This implementation is RISC-V microcontroller compatible, meaning the code can be compiled with RISC-V compiler, and memory-space requirements shouldn't collide with implementors' resources.
- The AI accelerator should implement the model, obtain acceleration compared to the software implementation running on the RISC-V core.
- Using the AI accelerator, competitors should match the accuracy of the software implemented model within a 10% window.
- Competitors are expected to analyze the model and its software implementation first, optionally follow the demo where the model is trained and tested from scratch. Then, an implementation where the network is realized on a specifically designed circuitry on RTL level should be planned.

AI accelerator workflow shall be as the following:

- The SoC core shall fetch and execute instructions to write configurations into control/status registers (CSR) of UART peripheral that is going to be used for the data stream, and AI accelerator's CSRs.

- After the configuration, UART-stream peripheral will transmit the data to be inferenced, and it shall be written to the desired accelerator memory address.
- The data input of the model shall be received from AI accelerator through its UART-stream interface.
- After the inference operation is complete, the SoC core shall be notified through an interrupt signal. Then, the core shall execute a pre-defined interrupt handling routine for it.
- Finally, the core shall write inference result through its UART interface.

A-2 Peripheral Registers

Peripherals and register definitions defined for the microcontroller are given in this appendix. The bus architecture and protocol for the connections of peripheral units shall be AXI-compliant. The definition of the base addresses of the peripheral units is left to the competitors.

GPIO

It is the peripheral unit responsible for reading and writing the values of a total of 32 input and output pins. 16 pins are fixed as input and 16 pins as output.

Offset	Name	Description	R/W
0x00	GPIO_IDR	Input data register. It holds the value of the 16-bit input signal in GPIO_IDR[15:0] bits. The value '0' is always present in the GPIO_IDR[31:16] bits.	RO
0x04	GPIO_ODR	Output data register. It transmits the 16-bit value written to the GPIO_ODR[15:0] bits to the output bits. The value written to bits GPIO_ODR[31:16] has no effect.	RW

Timer

32-bit timer peripheral. Its value increases or decreases based on the system clock frequency.

Offset	Name	Description	R/W
0x00	TIM_PRE	Prescaler register. The value of the counter register will be changed by dividing the clock frequency by the value written to the TIM_PRE register. For example, when TIM_PRE is '0', the counter (TIM_CNT) will increase/decrease at the system clock rate. In other words, it will change 1 in 1 period of the system clock. When TIM_PRE is '1', the counter will change 1 every 2 periods of the system clock. When TIM_PRE is "0xFFFFFFFF" the counter (TIM_CNT) will change 1 every 0x80000000 clock periods.	RW
0x04	TIM_AR	Auto-reload register. The counter register (TIM_CNT) will take the value 0 when it reaches the TIM_AR value and will continue its normal operation in the next period. For	RW

		example, if the TIM_AR register is 0x36, it will take the value 0 in the period after TIM_CNT reaches the value 0x36.	
0x08	TIM_CLR	Clear register. If the TIM_CLR[0] bit is '1', the counter register (TIM_CNT) value will be set to 0. Other bits are ineffective.	RW
0x0C	TIM_ENA	Enable register. If the TIM_EN[0] bit is '1', the counter will continue its normal operation. If the TIM_EN[0] bit is '0', the counter will retain its last value. If TIM_EN is '0' and TIM_CLR is '1', the counter register will be reset. Other bits are ineffective.	RW
0x10	TIM_MOD	Mode register. If the TIM_MOD[0] bit is '1' it will count up, if it is '0' it will count down.	RW
0x14	TIM_CNT	Timer counter register. The register where the counter is kept.	RO
0x18	TIM_EVN	Event register. The TIM_CNT register will increment by 1 each time it reaches the TIM_ARE value.	RO
0x1C	TIM_EVC	Event clear register. If the TIM_EVC[0] bit is '1', the event register (TIM_EVN) value will be set to 0. Other bits are ineffective.	RW

UART

Even though the register list below is applicable to the base UART, the UART-stream peripheral may contain additional registers depending on the design choices of the competitors. The programmable baud rate of the UART module shall support 1 Mbps data transfer speed and it shall support at least two different baud rates.

Offset	Name	Description	R/W
0x00	UART_CPB	Clock-per-bit register. It tells you how many clock pulses will be sampled for each bit in the UART packet. It is used to set the baud rate. For example, if there is a system clock at 48 MHz frequency, $48e6/9600 = 5000$ must be written to the UART_CPB register to communicate with a baud rate of 9600 bps. UART baud rate will be calculated as (system clock frequency)/UART_CPB. The UART peripheral to be designed should work this way.	RW
0x04	UART_STP	Stop-bit register. According to the UART_STP[1:0] value, the stop bit width is determined as follows: "00": Stop-bit 1 "01": Stop-bit 1.5 "1X": Stop-bit 2 Other bits are ineffective.	RW

0x08	UART_RDR	Read data register. It records 1 byte of data into UART_RDR[7:0] bits by the UART receiver. Other bits are ineffective. When data reception is completed, it sets the UART_CFG[1] bit to '1'.	RO
0x0C	UART_TDR	Transmit data register. It sends the data in the UART_TDR[7:0] bits to the other party when the UART_CFG[0] bit is set to '1'. When data sending is completed, it sets the UART_CFG[2] bit to '1'.	RW
0x10	UART_CFG	Configuration register. Its features are as follows: - UART_CFG[0]: Transmit start bit. It starts the transfer of data on UART_TDR register when asserted to '1'. When the transfer is complete, this bit should be deasserted to '0' by logic itself. - UART_CFG[1]: Data received bit. When data arrives, it is set to '1' by HW. It must be set to '0' by SW. - UART_CFG[2]: Transmit completed bit. When the data in UART_TDR is sent, it is set to '1' by HW. It must be set to '0' by SW.	RW

I2C Master

The SCL clock frequency of the I2C Master peripheral to communicate with the slave device will be at a fixed speed of 400 kHz. You can benefit from the documentation in the following GitHub repo about the I2C Master working principle:

https://github.com/mbaykenar/apis_anatolia/tree/main/VHDL_ile_FPGA_PROGRAMLA/MA/ders27

Offset	Name	Description	R/W
0x00	I2C_NBY	Number of bytes register. It refers to the number of bytes to be written to or read from the slave device by the I2C master peripheral. It can take a value between 1-4. If a value other than 1,2,3,4 is entered, it will be rounded to the nearest number. For example, if you write 0, it gets the value 1, if you write 25, it gets the value 4.	RW
0x04	I2C_ADR	Slave address register. 7-bit mode is supported. I2C_ADR[6:0] bits define the slave address. Other bits are ineffective.	RW
0x08	I2C_RDR	Read data register. It contains the data to be read from the slave device by the I2C master. It can save 1 to 4 bytes according to the I2C_NBY register value. When the I2C_CFG[2] bit is '1', it starts the receive process. When the receive operation is completed, it sets the I2C_CFG[3] bit to '1'. The first arriving data is written into low significant bytes. For example, if I2C_NBY is 2, the first read byte is	RO

		written to I2C_RDR[7:0] bits, and the second read byte is written to I2C_RDR[15:8] bits.	
0x0C	I2C_TDR	Transmit data register. It contains the data to be written to the slave device by the I2C master. It can transmit between 1 and 4 bytes according to the I2C_NBY register value. When the I2C_CFG[0] bit is '1', it performs the transfer. At the end of the transfer, it sets the I2C_CFG[1] bit to '1'. Transmission occurs starting from the low significant byte. For example, if I2C_NBY is 2, the data in I2C_TDR[7:0], then I2C_TDR[15:8] bits are transmitted.	RW
0x10	I2C_CFG	Configuration register. Since transmit and receive transfer cannot be done at the same time in the I2C master, hardware precautions must be taken if receive and transmit are enabled at the same time while designing the peripheral. For example, if I2C_CFG[0] and I2C_CFG[2] are '1' at the same time, transmit is made or other measures can be taken. Its features are as follows: - I2C_CFG[0]: Transmit enable bit. As long as it is '1', it sends the data in the I2C_TDR register in the amount of I2C_NBY bytes. - I2C_CFG[1]: Transmit completed bit. When data transmission is completed, it is set to '1' by HW. It must be set to '0' by SW. - I2C_CFG[2]: Receive enable bit. As long as it is '1', it starts the transfer to read the number of bytes in the I2C_NBY register from the slave device. - I2C_CFG[3]: Receive completed bit. When reading data from the slave device is completed, it is set to '1' by HW. It must be set to '0' by SW.	RW

QSPI Master

The QSPI Master peripheral will communicate with the non-volatile memory that will be used for the microcontroller program memory. An example NOR Flash (MT25QL256ABA8E12) datasheet with which the QSPI Master peripheral will communicate can be accessed from the links below:

Datasheet can be accessed from [this link](#).

Verilog Simulation Model (can be signed up with a student email address for access) can be accessed from [this link.https://www.infineon.com/dgdl/Infineon-S25FL128SS25FL256S_128_Mb_\(16_MB\)256_Mb_\(32_MB\)_3.0V_SPI_Flash_Memory-DataSheet-v20_00-EN.pdf?fileId=8ac78c8c7d0d8da4017d0ecfb6a64a17](https://www.infineon.com/dgdl/Infineon-S25FL128SS25FL256S_128_Mb_(16_MB)256_Mb_(32_MB)_3.0V_SPI_Flash_Memory-DataSheet-v20_00-EN.pdf?fileId=8ac78c8c7d0d8da4017d0ecfb6a64a17)

The QSPI Master peripheral shall support x1, x2 and x4 data width. Data transactions will support 256 bytes page write and page read. There will be 4-byte addressing mode support to cover all the flash area. SPI mode 0 will be supported (SCLK pin will stop at '0'

in IDLE state; CS pin will start sampling with the rising edge after falling to '0'). Dual-flash mode will not be supported. It will work as SDR (single data rate); DDR (double data rate) will not be supported. The following commands will be supported in QSPI Flash memory:

- READ
- DOR (Read Dual Out)
- QOR (Read Quad Out)
- PP (Page Program)
- QPP (Quad Page Program)
- SE (Sector Erase)
- READ_ID (Read Electronic Manufacturer Signature)
- RDSR1 (Read Status Register-1)
- RDSR2 (Read Status Register-2)
- RDCR (Read Configuration Register)
- WRR (Write Register – Status-1, Configuration-1)
- WRDI (Write Disable)
- WREN (Write Enable)
- CLSR (Clear Status Register-1)
- RESET (Software Reset)

To send these instructions to flash memory, the appropriate instruction value will be written to the "Instruction" bits in the QSPI_CCR register. Instruction values should be adjustable for software. If a competing team doesn't have access to the memory part specified, they are free to use any NOR flash memory as long as it supports all the specifications in this section. If this is not the case, it shall be inquired through the competition group for clarification. In any scenario where the part specified above isn't used, it shall be clarified in the reports and presentation.

In order to execute data operations, there is a single 32-bit data register in the design, which shall have an RX and a TX FIFO behind it. The FIFO specifications are left to the competitors' judgement, and the only restrictions on their design are the ones mentioned in the register definitions. QSPI Master's behavior under erroneous circumstances (e.g. data read when FIFO empty, or data write when FIFO full, etc.) shall be defined by the competitors.

Offset	Name	Description	R/W
0x00	QSPI_CCR	Communication configuration register. When a transfer occurs for writing or reading to flash memory, a value is written to this register. The QSPI peripheral controller initiates the transaction for reading/writing with the flash memory when a write is made to this register. - QSPI_CCR[7:0]: Instruction value. It contains the instruction value to be sent to flash memory. - QSPI_CCR[9:8]: Data mode:	RW

		- “00” Sending data - “01”x1 - “10”x2 - “11”x4 - QSPI_CCR[10]: - '0' Reading from flash memory - '1' Writing to flash memory - QSPI_CCR[15:11]: Number of dummy cycles. The flash memory may request a dummy cycle with READ, DOR, QOR or other commands (the datasheet should be examined and a decision is made). In this case, a dummy cycle is applied as much as the value in this field. - QSPI_CCR[23:16]: Data size to be written or read to flash memory. One more byte of data is written or read than the value written into this field. For example, if the WRR command is only written to Status Register-1 in the Flash memory, 0 is entered in this field, and if it is also written to the Configuration Register, 1 is entered. 255 is entered for the PP command. Reading is similar. - QSPI_CCR[24]: RESERVED. - QSPI_CCR[30:25]: Prescaler value. The value of the counter register will change by dividing the clock frequency by one more than the value entered in this field. For example, if '0' is entered in this field, SCLK will be at the system clock speed. When this field is '1', SCLK will be at half the system clock speed. - QSPI_CCR[31]: Clear status register - '0' No effect. - '1' Clear status register QSPI_STA.	
0x04	QSPI_ADR	Address register. It contains the address information to be read or written in flash memory. It is used as 4-byte in READ, DOR, QOR and PP command. That is, QSPI_ADR[31:0] bits are sent as the address.	RW
0x08	QSPI_DR	Data register. A page read from or written to the flash memory shall put the data into this register. There shall be one RX and one TX FIFO, each having a size of 64x32-bits connected to this register, to be able to process page program commands. The ordering of the data (LSB-first / MSB-first) processed from this register shall be determined by the competitors.	RW
0x0C	QSPI_STA	Status register.	RW

		- QSPI_STA[0]: When it is '1', the read or write transaction is completed. It is reset when '1' is written to the QSPI_CCR[31] bit. - QSPI_STA[1]: - '0' Not busy - '1' Busy. In other words, it will be '1' as long as a transaction continues with the flash memory. - QSPI_STA[4]: RX FIFO full. This field is asserted to 1 when data FIFO is full, and deasserted to 0 when it isn't. - QSPI_STA[5]: RX FIFO empty. This field is asserted to 1 when data FIFO is empty, and deasserted to 0 when it isn't. - QSPI_STA[6]: TX FIFO full. This field is asserted to 1 when data FIFO is full, and deasserted to 0 when it isn't. - QSPI_STA[7]: TX FIFO empty. This field is asserted to 1 when data FIFO is empty, and deasserted to 0 when it isn't. - QSPI_STA[11:8]: FIFO error. This field asserts the following values in given conditions: - '0000' No error, - '0001' RX FIFO read when empty (write 1 to clear), - '0010' TX FIFO write when full (write 1 to clear), - '{other values}' User Defined / Reserved. - The fields that aren't defined are User Defined / Reserved.	
0x10	QSPI_FCR	FIFO control register. - QSPI_FCR[0]: RX FIFO flush. Flushes the RX FIFO when asserted to 1, and auto-clears to 0. - QSPI_FCR[1]: TX FIFO flush. Flushes the TX FIFO when asserted to 1, and auto-clears to 0. - The fields that aren't defined are User Defined / Reserved.	RW

JTAG (Optional)

The JTAG port has been added to the specification and is optional in order to debug the CV32E40P processor core, which is the RISC-V based processor core of the microcontroller. The CV32E40P processor has a "debug" port and the processor can be debugged through this port. The processor's debug port can be accessed via a JTAG TAP controller within the microcontroller. Details are available in the links below. **(Competitors who successfully add this interface will be awarded +3 bonus points)**

CV32E40P processor core Debug & Trigger documentation page:

<https://docs.openhwgroup.org/projects/cv32e40p-user-manual/en/latest/debug.html#debug-support>

Open-source GitHub repo for processor debug access with JTAG TAP implementation:

<https://github.com/pulp-platform/riscv-dbg/tree/master>

A-3 Verification Methods

This section introduces different levels of and methodologies of verification to provide an insight regarding the verification aspects. Within the scope of this competition, it is not expected for competitors to have perfect verification flows with extensive plans or near-perfect coverages given the time constraints. However, the more these methods are used, the more robust the verification activities; hence, the scoring would be.

It is not expected to have a full-fledged UVM-based system verification environment. However, as all the AXI interfaces would have agents integrated on them for protocol-checking purposes, it would be much easier for teams to integrate UVM-based self-checking testbench environments as the VIPs would provide all transactions in a self-contained manner, making possible scoreboarding activities much easier.

For effective use of UVM, SVA, and coverage; **Metrics DSim simulator** with a non-commercial license is suggested until the official commercial licenses are provided. Also, **Verilator** recently announced full-support for UVM, which may be considered as an open-source option.

Name	Priority	Description
Verification Plan	Best Effort**	The outline of the verification activities to be held, and targets to call the verification activities complete.
Block Level Tests	Optional***	Singular tests for the IP blocks that are designed or obtained open-source to test the functional correctness of the corresponding IP blocks. These can be directed or randomized.
Protocol Checks	Mandatory*	Protocol checking is the active checking of the specific protocols throughout the simulation runs to catch any non-compliant behavior (e.g. X data while there is a valid handshake is a protocol violation). These are mostly set up as SystemVerilog assertions (SVA) and they come built-in within the commercial protocol VIPs. Thus, if an AXI interface has an AXI VIP agent connected to it (for example, as a passive monitor), the connected agent would report any non-compliant behavior in the forms of errors or fatals during simulation runtime.
Core Tests	Best Effort**	Individual tests of CV32E40P core using Spike ISS and software tests (C / assembly) to see if instruction

		traces match in type and order to ensure the correctness of the core. These tests should be self-checking, and can work either in online (using core trace interfaces during sim) or offline (using plaintext data processing post-sim) manner.
AI Accelerator Tests	Mandatory*	Singular tests for the AI accelerator block that drive reference sound data (examples can be found in TFLite Micro repository, and 1000-milliseconds mono-channel WAV files can be converted to test data as well) from memory to the accelerator core, and checks the outputs. These tests can be directed or randomized, but they should be self-checking without requiring manual examination.
System-Level Tests	Mandatory*	The tests that are held on system level where custom C codes are written to initiate and use one or multiple peripherals integrated on the SoC to check the block integration integrity on SoC level. These tests can be directed or randomized, but they should be self-checking without requiring manual examination.
Code Coverage	Optional***	These metrics are auto-generated through simulators, and indicate how much of the RTL code executed in different metrics (lines, toggling activity, branches taken). These coverages can be merged throughout multiple simulations. Low code coverage indicates either the design not being tested enough, or there is unused code in the design.
Functional Coverage	Optional***	These metrics are user-defined and obtained through simulations by the simulators. These can be in the form of assertions (SVA) or coverage (SystemVerilog coverage). These coverages can be merged throughout multiple simulations. These metrics quantify the design's intent, and verification should always target 100% with the defined functional coverage points.

***Mandatory:** These are required to be eligible for the awards.

****Best Effort:** These have their own scoring within the Design & Verification scores.

*****Optional:** These help you achieve full points from the verification activities.

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